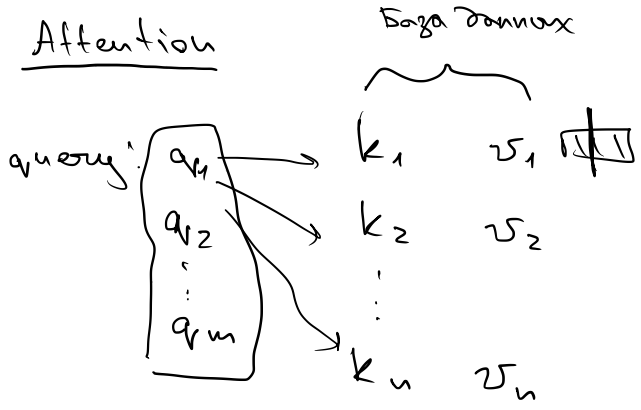
Attention



$$Att(q,k,v) = f(q) = \sum_{i=1}^{n} \frac{e^{\alpha(q,k_i)}}{\sum_{j=1}^{n} e^{\alpha(q,k_j)}} \cdot v_i = \sum_{i=1}^{n} P(k_i|q) \cdot v_i$$

$$\alpha(q,k_i) = \langle q, k_i \rangle / \sqrt{d}$$

$$Att(q,k,v) = W_O [\, head_1 \,|\, \dots \,|\, head_H \,]$$

- Cross-Attention vs Self-Attention

$X_1 \in \mathbb{R}^{n \times d}$

$X_2 \in \mathbb{R}^{m \times d}$

$X \in \mathbb{R}^{n \times d}$

Q, K, V

$Q = X W_Q, \quad W_Q \in \mathbb{R}^{d \times d}$

$K =$

$V =$

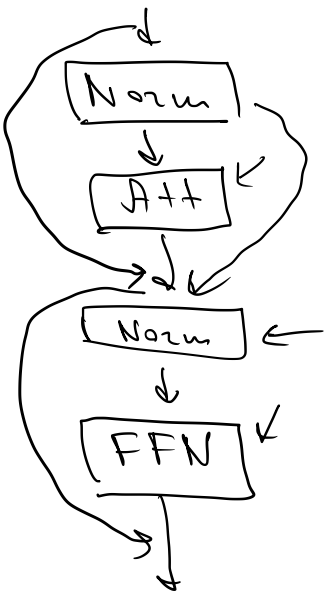
- Bidirectional vs Causal

$t_1 \dots t_n$

$$\begin{array}{c|cccc} t_1 & 1 & 0 & \dots & 0 \\ \vdots & 1 & 1 & \dots & 0 \\ t_n & 1 & 1 & \dots & 1 \end{array}$$

Transformer

$x \leftarrow$ residual stream



$FFN(x) = W_2\, \sigma(W_1 x)$

Prenorm

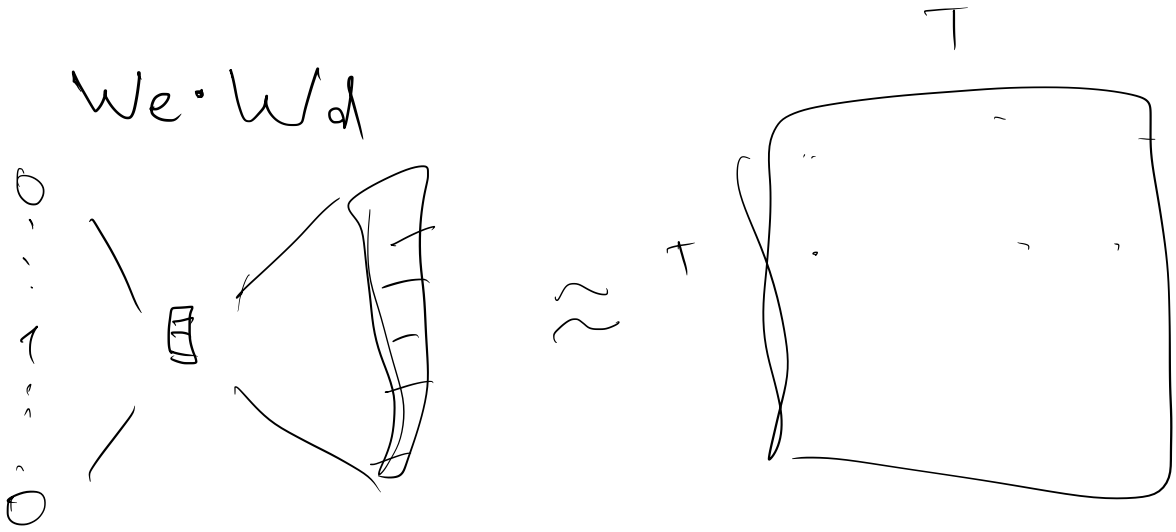
знач

ключ

LayerNorm

RmsNorm

$ABCA \to B$

decoder

$W_e \cdot W_d$